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## About

I am currently an Assistant Professor at AI Thrust, Information Hub, The Hong Kong University of Science and Technology (Guangzhou). My main research focus is quantum information theory, with topics including quantum causal inference, quantum data compression, hypothesis testing and benchmarks. I also have interest in quantum machine learning and circuit design.

Google scholar link: <https://scholar.google.com/citations?user=usrPjHoAAAAJ>

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## Expertise

- Quantum information theory
- Higher-order quantum operations
- Quantum benchmarks
- Quantum causal inference
- Tensor networks
- Classical machine learning and algorithms

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## Awards

Dec. 2021 Winner of 2021 Hong Kong Institution of Science Young Scientist Award

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## Positions

- Mar. 2025 to now Assistant professor, Thrust of Artificial Intelligence, Information Hub, The Hong Kong University of Science and Technology (Guangzhou), Guangdong, China.
- Feb. 2023 to Jan. 2025 Post-doctoral fellow, Centre for Quantum Technologies, National University of Singapore.
- Mar. 2022 to Jan. 2023 Post-doctoral fellow, Department of Computer Science, the University of Hong Kong.
- Apr. 2021 to Feb. 2022 Research assistant, Department of Computer Science, the University of Hong Kong.

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## Education

- Sep. 2016 to Mar. 2021 PhD, Department of Computer Science, the University of Hong Kong. Supervised by Prof. Giulio Chiribella.
- Aug. 2012 to Jul. 2016 BEng in Yao Class, Institute for Interdisciplinary Information Sciences (IIIS), Tsinghua University. Yao Class is a pilot class on computer science directed by the respected Turing Award winner, Andrew Yao.

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## Publications

Mingxuan Liu, Valerio Scarani, **Ge Bai**, “Proper and Improper Mixed States Serve as Different Prior Beliefs for Quantum State Retrodiction.” *Physical Review Letters*, 136.6 (2026), 060203.

**Ge Bai**, Francesco Buscemi, Valerio Scarani, “Quantum Bayes’ rule and Petz transpose map from the minimum change principle.” *Physical Review Letters*, 135.9 (2025): 090203.

**Ge Bai**, Dominik Šafránek, Joseph Schindler, Francesco Buscemi, Valerio Scarani, “Observational Entropy with General Quantum Priors.” *Quantum* 8, 1524 (2024).

**Ge Bai**, Iman Marvian, “Synthesis of Energy-Conserving Quantum Circuits with XY interaction.” *Quantum Science and Technology* 9, 045049 (2024).

Ya-Dong Wu, Yan Zhu, **Ge Bai**, Yuexuan Wang, Giulio Chiribella, “Quantum Similarity Testing with Convolutional Neural Networks.” *Physical Review Letters* 130, 210601 (2023).

Fei Shi, **Ge Bai**, Xiande Zhang, Qi Zhao, Giulio Chiribella, “Graph-theoretic characterization of unextendible product bases.” *Physical Review Research* 5, 033144 (2023).

Yan Zhu, **Ge Bai**, Yuexuan Wang, Tongyang Li, Giulio Chiribella. “Quantum autoencoders for communication-efficient quantum cloud computing.” *Quantum Machine Intelligence*, 5(2):27, (2023).

**Ge Bai**, Ya-Dong Wu, Yan Zhu, Masahito Hayashi, Giulio Chiribella. “Quantum causal unravelling.” *npj Quantum Information* 8.69 (2022).

Yan Zhu, Ya-Dong Wu, **Ge Bai**, Yuexuan Wang, Giulio Chiribella. “Flexible learning of quantum states with generative query neural networks.” *Nature Communications* 13, 6222 (2022).

Ya-Dong Wu, **Ge Bai**, Giulio Chiribella, Nana Liu. “Efficient verification of continuous-variable quantum states and devices without assuming identical and independent preparation.” *Physical Review Letters* 126.24 (2021): 240503

**Ge Bai**, Yuxiang Yang, Giulio Chiribella. “Quantum compression of tensor network states.” *New Journal of Physics* (2020): 043015.

**Ge Bai**, Giulio Chiribella. “Test one to test many: a unified approach to quantum benchmarks.” *Physical Review Letters* 120.15 (2018): 150502. **Editors’ Suggestion.**

Yuxiang Yang, **Ge Bai**, Giulio Chiribella, Masahito Hayashi, “Compression for quantum population coding.” *IEEE Transactions on Information Theory* (2018).

Xiao Yuan, **Ge Bai**, Tianyi Peng, Xiongfeng Ma. “Quantum uncertainty relation using coherence.” *Physical Review A* 96.3 (2017): 032313.

**Ge Bai**, Ivan Damgård, Claudio Orlandi, Yu Xia, “Non-interactive verifiable secret sharing for monotone circuits”, *International Conference on Cryptology in Africa*. Springer International Publishing, 2016.

**Ge Bai**, Hansi Mou, Yinhong Hou, Yongqiang Lyu, Weikang Yang, "Android power management and analyses of power consumption in an Android smartphone", *2013 IEEE International Conference on High Performance Computing and Communications & 2013 IEEE International Conference on Embedded and Ubiquitous Computing*. IEEE, 2013.

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## Preprints

Chengkai Zhu, Ziao Tang, Guocheng Zhen, Yinan Li, **Ge Bai**, Xin Wang, "Simulation of Adjoints and Petz Recovery Maps for Unknown Quantum Channels." arXiv:2602.05828 (2026).

Mingxuan Liu, **Ge Bai**, Valerio Scarani, "Unifying Quantum Smoothing Theories with Extended Retrodiction." arXiv:2510.08447 (2025).

**Ge Bai**, "Bayesian retrodiction of quantum supermaps." arXiv:2408.07885 (2024).